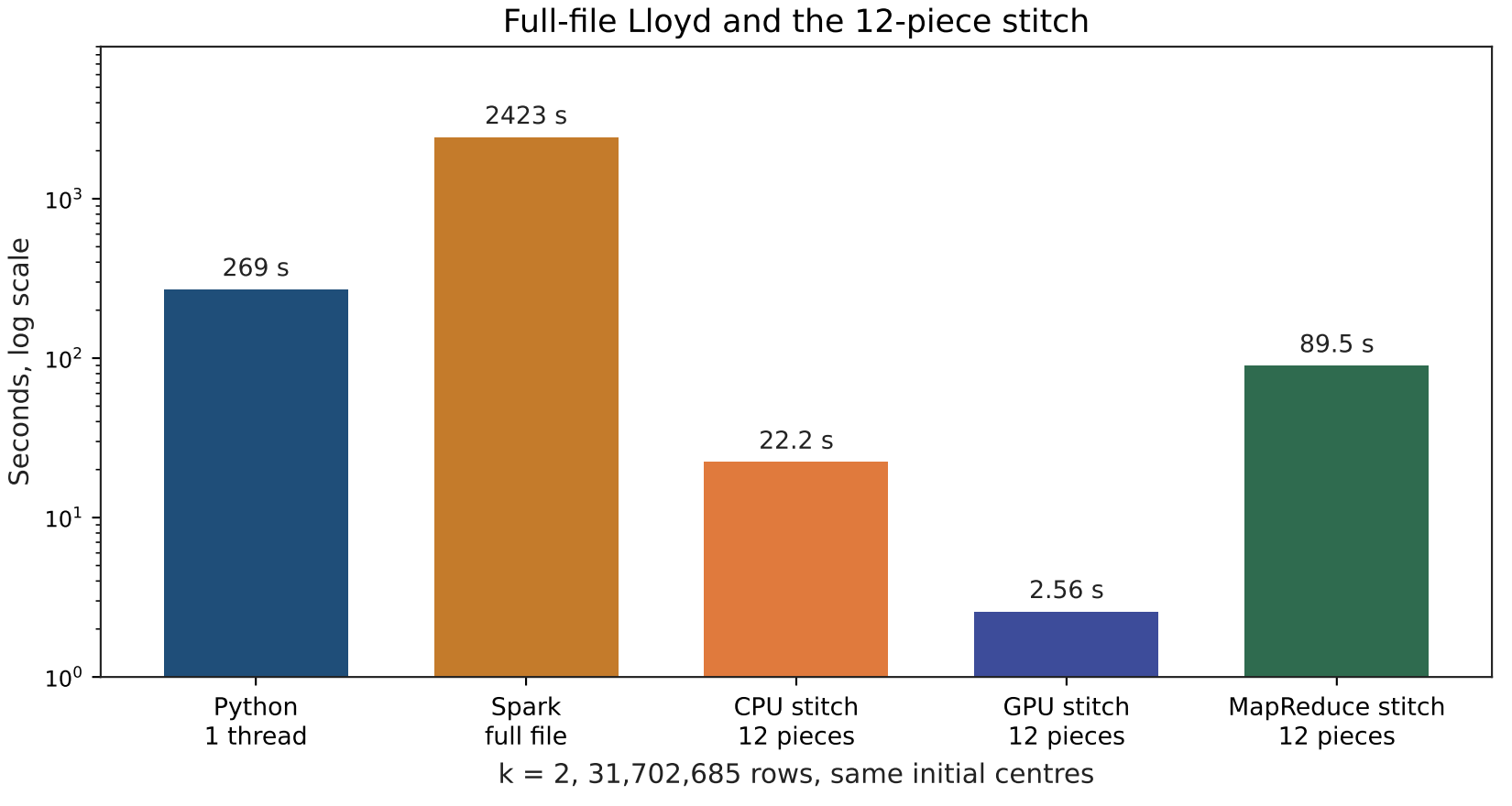

Python is 1 thread, 86 rounds on the whole file. Spark full file is local[12], also 86 rounds.

Stitch: each of 12 pieces clusters alone, then Lloyd on the 24 local centres. CPU is Spark local[12].

MapReduce is one YARN job on this PC, 8 vcores. Putting the pieces on HDFS, 18.0 s, is not in that bar.

GPU stitch is the Lloyd time on the RTX 3070. Copy onto the device, 0.72 s, is not in that bar.

CPU, GPU, and MapReduce stitch reach the same centres. They differ from the full-file centres by 1.489.